

MONTHLY INTERNSHIP REPORT

DeepInfinity — August 2026

Internship Progress & Task Summary

Organization	DeepInfinity
Reporting Period	August 2026
Area of Work	.NET, WPF, Azure Functions, PHP, Web Application Development
Report Type	Monthly Internship Report

1. Executive Summary

During August 2026, the internship work focused on understanding WPF and .NET desktop application development, setting up the local development environment, building and consuming a locally hosted Azure Function, and extending the application with file/image processing and audio transcription. The application was subsequently converted toward a web-based implementation using PHP and .NET while retaining the core functionality.

2. Task-wise Work Completed

Task 1 — WPF Fundamentals and Environment Setup

- Learned the fundamentals of Windows Presentation Foundation (WPF) and its role in building Windows desktop applications using .NET.
- Understood the separation between XAML for user-interface design and C# for application logic and event handling.
- Explored common WPF controls and layouts used to build desktop interfaces.
- Installed and configured Visual Studio for WPF development.
- Created and explored basic WPF applications to become familiar with the project structure and development workflow.

Task 2 — Local Azure Function and WPF Login Integration

- Created a sample Azure Function with an HTTP trigger for login functionality.
- Configured and executed the Azure Function locally without deploying it to Azure.
- Created a WPF login form containing username and password fields.
- Connected the WPF application to the locally running Azure Function through an HTTP request.
- Handled the login response and displayed the result in the WPF application.
- Used JSON-based request/response communication between the client and backend.

Task 3 — Image/File Content Extraction and Audio Transcription

- Extended the application after login with a form for uploading an image or file.
- Implemented the flow for sending uploaded content for processing and displaying the extracted/read content.

- Added a separate audio-upload form.
- Implemented the audio processing flow to generate and display a speech-to-text transcript.
- Integrated the new functionality into the existing application while maintaining the login flow.

Task 4 — Conversion to PHP and .NET Web Pages

- Converted the existing application from the WPF-oriented interface toward a web-based implementation using PHP and .NET.
- Preserved the core application workflow, including login, file/image processing, and audio transcription.
- Separated the web interface from backend processing and API communication.
- Configured the application to support local development and testing.
- Updated the project structure and code to support the new web-based implementation.

3. Technologies and Tools Used

Technology / Tool	Purpose	Usage During Internship
.NET / C#	Application and backend development	WPF application and backend/API functionality
WPF	Windows desktop UI development	Login form and desktop application interface
XAML	UI design in WPF	Layouts, controls, and form design
Azure Functions	Serverless backend/API	Local HTTP-triggered login function and application processing
PHP	Web application pages	Conversion of the application toward a web-based interface
ASP.NET / .NET	Web/backend functionality	Backend/API integration for the web implementation
HTTP / JSON	Client-server communication	Communication between frontend applications and backend services
Visual Studio	Development environment	Local development, debugging, and project setup

4. Overall Application Flow

1. User opens the application.
2. User enters login credentials in the login interface.
3. The client sends the credentials to the locally running backend/Azure Function.
4. The backend validates the credentials and returns a response.
5. After successful login, the user can access file/image processing and audio transcription features.
6. For an uploaded image or file, the content is processed and the extracted content is displayed.
7. For an uploaded audio file, speech is converted into text and the transcript is displayed.
8. The application was then adapted toward a PHP and .NET web-based implementation while retaining the main workflow.

5. Key Learning Outcomes

- Understood the basic architecture and workflow of WPF applications.
- Gained practical experience with XAML-based UI development and C# event handling.
- Learned how a client application communicates with a backend through HTTP APIs.
- Learned the basics of creating and running Azure Functions locally.
- Worked with JSON request and response data.
- Understood the application flow for file upload, content extraction, OCR-related processing, and audio transcription.
- Gained exposure to converting a desktop-oriented application into a web-based architecture using PHP and .NET.
- Improved practical experience with local development, integration, testing, and debugging.

6. Current Status

The assigned tasks for the reporting period were implemented and the updated code was maintained in the project repository. The application was tested in a local development environment as required by the assigned tasks.

7. Next Steps

- Continue development based on the mentor's next assignment.
- Refine the PHP and .NET implementation based on code review feedback.
- Improve validation, error handling, and user-interface behavior where required.
- Further understand the architecture and individual components so the implementation can be explained and maintained independently.